

Consumption

The Permanent Income Hypothesis under rational expectations

$$\max_{\{c_t, a_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

$$u' > 0$$

$$u'' < 0$$

$$c_t + \frac{a_{t+1}}{1+r_t} = y_t + a_t$$

$$\text{uBg condition} \quad \lim_{T \rightarrow \infty} \frac{a_{T+1}}{\prod_{t=0}^{T-1} (1+r_t)} \geq 0$$

a_0 given

FOCs give Euler equation

$$u'(c_t) = \beta (1+r_t) u'(c_{t+1}) \quad \forall t$$

also have ITBC

$$\underbrace{\sum_{j=0}^{\infty} \frac{1}{\prod_{k=0}^{j-1} (1+r_{t+k})}}_{\text{life-time consumption}} c_{t+j} = \underbrace{\sum_{j=0}^{\infty} \frac{1}{\prod_{k=0}^{j-1} (1+r_{t+k})} y_{t+j}}_{\text{life-time income}} + a_t$$

life-time wealth

Euler equation, consumption smoothing, and intertemporal substitution

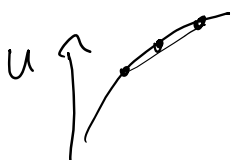
Assume $\beta(1+r_t) = 1$

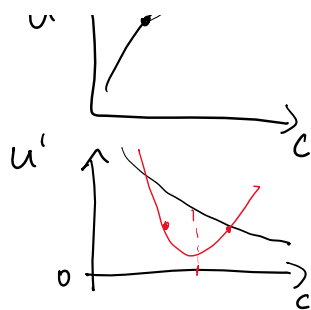
$$\Rightarrow u'' < 0 \Rightarrow$$

$$u'(c_t) = u'(c_{t+1})$$

$$c_t = c_{t+1}$$

Consumption smoothing





Consider constant relative risk aversion

$$u(c) = \frac{c^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}} \quad \sigma > 0$$

σ : intertemporal elasticity of substitution

$\frac{1}{\sigma}$: coefficient of relative risk aversion

$$\frac{1}{\sigma} := - \frac{u''(c)c}{u'(c)}$$

$$u'(c) = c^{-\frac{1}{\sigma}}$$

$$EE \Rightarrow c_t^{-\frac{1}{\sigma}} = \beta(1+r_t) c_{t+1}^{-\frac{1}{\sigma}}$$

$$\frac{c_{t+1}}{c_t} = [\beta(1+r_t)]^{\sigma}$$

$$- \beta(1+r_t) = 1 \Rightarrow \frac{c_{t+1}}{c_t} = 1$$

$$- \beta(1+r_t) > 1 \Rightarrow \frac{c_{t+1}}{c_t} > 1$$

$$- \beta(1+r_t) < 1 \Rightarrow \frac{c_{t+1}}{c_t} < 1$$

$$\beta = \frac{1}{1+\rho}$$

β : discount factor

ρ : rate of time preference

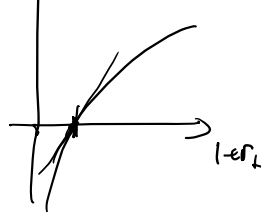
$$\frac{1+r_t}{1+\rho} > 1$$

$$\Leftrightarrow r_t > \rho$$

$$\begin{aligned} \text{EE} \\ \Rightarrow \quad \ln \frac{C_{t+1}}{C_t} &= \sigma \cdot \ln(1+r_t) + \sigma \ln \beta \end{aligned}$$

$$\begin{aligned} \text{def. of IES} \quad \rightarrow \quad \frac{\partial \ln \frac{C_{t+1}}{C_t}}{\partial \ln(1+r_t)} &= \sigma \quad \text{non-durable perishable goods} \\ \text{intertemporal elasticity of substitution} \\ \text{Estimates: } \sigma &\in [0.1, 0.5] \end{aligned}$$

$$\Delta \frac{C_{t+1} - C_t}{C_t} = \sigma \cdot \Delta r_t$$

$\Delta \ln(1+r_t) \approx \Delta r_t$


The marginal propensity to consume in the P_{IT} model

$$\text{mpc} := \frac{\partial C}{\partial y_t} \quad \text{notice: not logs}$$

$$\underline{R} = 1+r = \text{const}$$

ITBC:

$$\sum_{i=0}^{\infty} \frac{1}{R^i} C_{t+i} = a_t + \sum_{i=0}^{\infty} \frac{1}{R^i} y_{t+i}$$

EE:

$$- \ln(1+r)^{\sigma} = \ln(1+r)^{\sigma} \cdot C_t$$

EE:

$$C_{t+i} = (R\beta)^i C_{t+i-1} = ((R\beta)^i)^i C_t$$

$$\Rightarrow C_t = (1 - \bar{R}^{-1}(R\beta)^{\sigma}) \left[a_t + \sum_{i=0}^{\infty} \underbrace{\frac{1}{R^i}}_{\uparrow} \underbrace{y_{t+i}}_{\uparrow} \right]$$

consider $i=0$

$$mpc = \frac{\partial C_t}{\partial y_t} = 1 - \bar{R}^{-1}(R\beta)^{\sigma}$$

$$\begin{aligned} R\beta &\approx 1 \\ \frac{1+r}{1+g} &= 1 \quad r=g \\ R\beta &\approx 1 \quad \approx 1 - \frac{1}{R} = \frac{1+r}{1+g} - \frac{1}{1+g} \approx r \end{aligned}$$

in the PII model, the mpc is small!

$$\approx 0.03$$

One substitute

$$y_t^c = y_t - y_t^p$$

$$y_t = \underbrace{y_t^p}_{\substack{\uparrow \\ \text{permanent or} \\ \text{average income}}} + y_t^c \quad \text{current income}$$

$$\begin{aligned} C_t &= (1 - \bar{R}^{-1}(R\beta)^{\sigma}) \left(a_t + \sum_{i=0}^{\infty} \frac{1}{R^i} (y_t^p + y_t^c) \right) \\ &= (1 - \bar{R}^{-1}(R\beta)^{\sigma}) \left(a_t + y_t^p \underbrace{\frac{1}{1-R}}_r + \sum_{i=0}^{\infty} \frac{1}{R^i} y_t^c \right) \end{aligned}$$

$$mpc = \frac{\partial C_t}{\partial a_t}$$

$$\frac{1}{1-R} = \frac{1}{1-(1+r)^{-1}} = \frac{1+r}{r}$$

$$\frac{\partial C_t}{\partial y_t^p} = (1 - \bar{R}^{-1} \underbrace{(R\beta)^{\sigma}}_{\approx 1}) \frac{1+r}{r}$$

$$= \frac{1}{r}(1+r-1) = 1$$

Summary:

$$\frac{\partial G}{\partial y_t} = \frac{\partial G}{\partial y_t^c} \approx \phi \quad \text{for transitory change in } y$$

$$\frac{\partial G}{\partial y^p} \approx 1 \quad \text{for permanent change in } y$$

the MPC depends on the persistence of the income change

Friedman 1957 book

Theory of the consumption function.

$$\begin{aligned} \frac{C_t}{Y_t} &= \phi & \Rightarrow C_t &= \phi Y_t \\ & & \Rightarrow \frac{\partial C_t}{\partial Y_t} &= \phi \end{aligned}$$

The random walk hypothesis and Rational Expectations errors

consider $\max_{\{C_t, a_{t+1}\}_{t=0}^{\infty}} E_0 \sum_{t=0}^{\infty} \beta^t u(C_t)$

s.t. $C_t + \frac{a_{t+1}}{1+r} = y_t + a_t$

$$\lim_{T \rightarrow \infty} \frac{a_{T+1}}{\prod_{t=0}^T (1+r_t)} \geq 0$$

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^T (1+r_t) \geq 0$$

a_0 given

$$\Rightarrow u'(c_t) = \beta(1+r_t) E_t[u'(c_{t+1})]$$

Hall (1978) showed the consumption approximately follows a random walk

Digression: Random Walk and RE errors

S_t : generic variable

$$S_{t+1} = S_t + \varepsilon_{t+1}$$

ε_t is white noise: $E[\varepsilon_t] = 0$, $E[\varepsilon_t^2] = \sigma_\varepsilon^2$

$$E[\varepsilon_t \varepsilon_\tau] = 0 \text{ for } t \neq \tau$$

$$\text{Consider } E_t[S_{t+1}] = E[S_{t+1} | \mathcal{I}_t]$$

$$\mathcal{I}_t = \{c_{t-j}, y_{t-j}, a_{t-j}\}_{j=0}^{\infty}$$

RE error:

$$\varepsilon_{t+1} := S_{t+1} - E_t[S_{t+1}]$$

$$(1) E_t[\varepsilon_{t+1}] = 0$$

$$E_t[S_{t+1} - E_t[S_{t+1}]] = E_t[S_{t+1}] - E_t[S_{t+1}] = 0$$

take x_τ $\tau \leq t$ i.e. $x_\tau \in \mathcal{I}_t$

$$(2) E_t[x_\tau \varepsilon_{t+1}] = x_\tau E_t[\varepsilon_{t+1}] = 0$$

in data
often not
true

$$(3) \quad \varepsilon_{t+1} = \alpha + \beta x_t + u_{t+1} \quad (2)$$

$$\beta = \frac{\text{cov}(\varepsilon_{t+1}, x_t)}{V[x_t]} = \frac{E[\varepsilon_{t+1} x_t]}{V[x_t]} = \frac{E[\overbrace{E_t[\varepsilon_{t+1} x_t]}^0]}{V[x_t]}$$

$$= 0$$

Random Walk hypothesis

$$u'(c_{t+1}) = E_t[u'(c_{t+1})] + \underbrace{\varepsilon_{t+1}}_{\text{def of RE error}}$$

EE
 $\Rightarrow$

$$u'(c_t) = \beta(1+r) u'(c_{t+1}) - \beta(1+r) \varepsilon_{t+1}$$

$\beta(1+r) \approx 1$
 $\Rightarrow$

$$u'(c_{t+1}) \approx u'(c_t) + \varepsilon_{t+1}$$

MU approx follows a random walk

Take a first order approx (Taylor) of MU around c , steady state level consumption

$$\begin{aligned} u'(c_t) &= u'(c) + u''(c) \cdot (c_t - c) \\ &= u'(c) + u''(c) \cdot c \left(\frac{c_t - c}{c} \right) \\ &= u'(c) + u''(c) c (\ln c_t - \ln c) \end{aligned}$$

$$\Rightarrow \ln c_{t+1} = \ln c_t + \eta_{t+1}$$

where $\eta_t := \frac{1}{cu''(c)} \varepsilon_{t+1}$

$$E_t[\ln C_{t+1}] = \ln C_t$$

Hall tests this prediction

$$\ln C_{t+1} = \ln C_t + \gamma X_t + \underline{\eta_{t+1}}$$

$H_0: \gamma = 0$ X_t : past income, stock prices, etc,

Statistical Tests of the PIH

Shea 1995 : Union contracts

predicted wage change causes
consumption change $\nless$ PIH

Hsieh $\approx$ 1999 , Kueang (2016)

Johnson, Parker, Souleles (2006)

study Economic Growth and Tax Relief
Reconciliation Act of 2001

tax rebates $\approx$ 300-600 \$ randomized receipt time

$$C_{i,t} - C_{i,t+1} = \beta R_{i,t} + \text{controls}_{i,t} + u_{i,t}$$

$\uparrow$
rebate
 ≈ 0 under PIH

Kaplan-
Violante
2014

CRRA $\rightarrow \beta \approx 9$ for transitory income change
 $\beta = 0$ for anticipated income changes

$$\hat{\beta} \approx 0.2 - 0.3 \quad \hookrightarrow \text{PIH}$$

Extensions of the PIH model

i) Precautionary savings

EE with $\beta(1+r) = 1$

$$u'(c_t) = E_t[u'(c_{t+1})]$$

2nd order Taylor exp. around c_t

$$u'(c_{t+1}) = u'(c_t) + u''(c_t) \cdot (c_{t+1} - c_t) + \frac{u'''(c_t)}{2} (c_{t+1} - c_t)^2$$

$$E_t\left[\frac{c_{t+1} - c_t}{c_t}\right] = - \underbrace{\frac{u'''(c_t) c_t}{2 u''(c_t)}}_{u'' < 0} \underbrace{E_t\left[\left(\frac{c_{t+1} - c_t}{c_t}\right)^2\right]}_{\approx \psi}$$

sign determined
by $u'''(c_t)$
↑
prudence
motive

coefficient of
relative
prudence

Kimball 1990

Assume $u''' > 0$

$$\text{then } E_t\left[\frac{c_{t+1} - c_t}{c_t}\right] > 0$$

This is known as precautionary savings

Quadratic utility

$$u(c_t) = b_1 c_t - \frac{1}{2} b_2 c_t^2$$

$$u' = b_1 - b_2 c_t$$

$$u'' = -b_2$$

$$u''' = 0 \Rightarrow \text{no prudence}$$

CRRA

$$u(c_t) = \frac{c_t^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}}$$

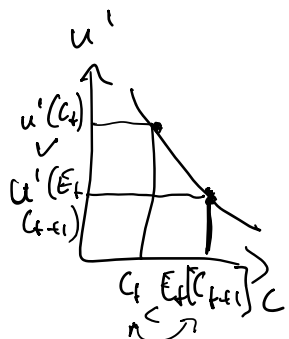
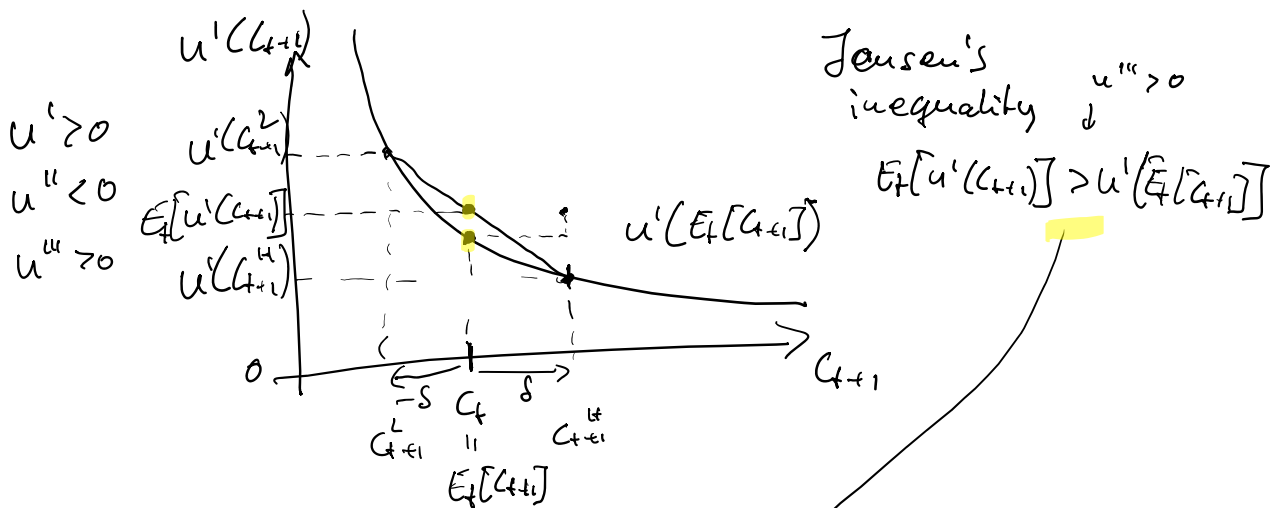
$$u' = c_t^{-\frac{1}{\sigma}}$$

$$u'' = -\frac{1}{\sigma} c_t^{-\frac{1}{\sigma}-1}$$

$$u''' = \frac{1}{\sigma} \left(\frac{1}{\sigma} + 1 \right) c_t^{-\frac{1}{\sigma}-2} > 0 \Rightarrow \text{prudence}$$

Illustration

$$u'(c_t) = E_t[u'(c_{t+1})]$$



$$u'(c_t) = E_t[u'(c_{t+1})] > u'(E_t[c_{t+1}])$$

$$\boxed{c_t < E_t[c_{t+1}]}$$

$$|(u')'|$$

$$u'' < 0$$

Risk aversion vs. Prudence

linear utility $u = c \Rightarrow E[u] = E[c]$

quadratic utility $u = a - b c^2 \Rightarrow E[u] = a E[c] - b E[c^2]$
 $= a(p c_L + (1-p) c_H) - b[p c_L^2 + (1-p) c_H^2]$

2) Borrowing Constraint

$$a_{t+1} \geq \underline{a} \quad \underline{a} \leq 0$$

(i) ad-hoc

(ii) natural borrowing limit

$$\forall c_t = 0$$

$$a_{t+1} + \sum_{j=0}^{\infty} \frac{1}{\prod_{k=0}^{j-1} (1+r_{t+1+k})} y_{t+1+j} = \sum_{j=0}^{\infty} \frac{1}{\prod_{k=0}^{j-1} (1+r_{t+1+k})} c_{t+1+j} \geq 0$$

$$a_{t+1} \geq - \sum_{k=0}^{\infty} \frac{1}{\prod_{k=0}^{j-1} (1+r_{t+1+k})} y_{t+1+k}$$

r const

$y_{\min}$

$$a_{t+1} \geq - y_{\min} \sum_{i=0}^{\infty} \left(\frac{1}{1+r} \right)^i = - \frac{1+r}{r} y_{\min}$$

$$\max_{\{c_t, a_{t+1}\}_{t=0}^{\infty}} E_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$$

$$\text{s.t.} \quad c_t + \frac{a_{t+1}}{1+r} = y_t + a_t$$

$$a_{t+1} \geq \underline{a}$$

←

$$\lim_{T \rightarrow \infty} \frac{a_{T+1}}{\prod_{t=0}^T (1+r_t)} \geq 0$$

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left(u(c_t) + \lambda_t \left(y_t + a_t - c_t - \frac{a_{t+1}}{1+r} \right) + \mu_t (a_{t+1} - \underline{a}) \right)$$

FOC:

$$u'(c_t) = \lambda_t$$

$$\beta E_t[\lambda_{t+1}] - \lambda_t \frac{1}{1+r} + \mu_t = 0$$

Rearrange:

$$u'(c_t) = \beta(1+r) E_t[u'(c_{t+1})] + \mu_t(1+r)$$

complementary slackness

$$\mu_t (a_{t+1} - \underline{a}) = 0$$

$$\mu_t \geq 0$$

Case 1: $c_{t+1} > \underline{a}$, $\mu_t = 0$

$$u'(c_t) = \beta(1+r) E_t[u'(c_{t+1})]$$

Case 2: constrained $c_{t+1} = \underline{a} \Rightarrow \mu_t \geq 0$

$$u'(c_t) = \beta(1+r) E_t[u'(c_{t+1})] + \underline{(1+r)\mu_t} \leftarrow$$

$$> \underline{\beta(1+r) E_t[u'(c_{t+1})]}$$

$$c_t + \frac{a_{t+1}}{1+r} = y_t + a_t$$

$$a_{t+1} = \underline{a}$$

$$c_t = y_t + a_t - \frac{a}{1+r}$$

$$mpc = \frac{\partial c_t}{\partial y_t} = 1$$

HANK

Borrowing constraints induce a saving motive

$$\beta(1+r) = 1$$

$$u'(c_t) \geq E_t[u'(c_{t+1})] \quad \forall t$$

Supermartingale

under conditions the supermartingale convergence thm. applies:

$$u'(c_t) \xrightarrow{a.s.} u'(c)$$

$$P\left(\lim_{t \rightarrow \infty} u'(c_t) = u'(c)\right) = 1$$

$$a_{t+1} \xrightarrow{a.s.} a$$

c, a grow to infinity

This called self-insurance

Sometimes (to distinguish from precautionary savings) use quadratic utility

$$b_1 - b_2 c_t \geq E_t[b_1 - \frac{b_2}{2} c_{t+1}]$$

$$\Rightarrow a \leq E_t[a_{t+1}]$$